

BASS LOW END Technical Manual

Bass, sub, foundation, and low-end transformation

Build weight and depth while retaining definition, placement, and controlled output.

Included products: MARIANA, MONOLITH, Depth Boost Pro, NTRPL

Routing authority: Bass route: MARIANA -> NTRPL -> MONOLITH. Depth Boost Pro is an optional focused stage.

Canon baseline

Normalized control 1 equals 0.809016994 (PHI/2). Standard knob travel is 7 o'clock to 5 o'clock. Factory programs must be clickless and must honor protected I/O, Mix, Power, and ceiling controls. Smart Position remains the placement authority.

MARIANA

Sub and foundation processor with depth, sub pressure, kick definition, target mode, parallel mix, and an optional kick-key sidechain that ducks only MARIANA's added low-end layer.

Declared product: MARIANA | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Low End Source | **Channel mode:** not detected | **Dual-mono link:** not detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
- No restricted DSP marker detected by the current source scan

Parameter surface

ID	Display name	Type
power	Power	Bool
depth	Depth	Float
pressure	Sub Pressure	Float
foundation	Foundation	Float
kick_define	Kick Define	Float
mode	Target Mode	Choice
input	Input	Float
mix	Mix	Float
gainmatch	Gain Match	Bool
sidechain_enable	Sidechain Key	Bool
sidechain_duck	Sidechain Duck	Float
sidechain_recovery	Key Recovery	Float

Artwork authority

assets/Mariana.png

Meter and visualizer semantics

Display	What it measures	How to read it
DEPTH	Current trench-depth control state	Higher values extend the added low-end field deeper.
PRESSURE	Current sub-pressure control state	Higher values add more low-frequency pressure.
FOUNDATION	Current sustained-foundation control state	Higher values add more continuous low-end support.
IMPACT	Current kick-definition control state	Higher values preserve more front-edge definition.
LOW ENERGY	Detected low-frequency program energy	Shows the bass energy entering MARIANA's field.
BEDROCK	Detected sustained foundation energy	Shows the slower body underneath transient bass events.
KEY	Detected 35-160 Hz sidechain energy	Moves only when KEY is enabled and a host sidechain is routed.
DUCK GR	Reduction of MARIANA's added low-end layer	More activity means the external key is clearing more generated pressure; dry bass remains audible.

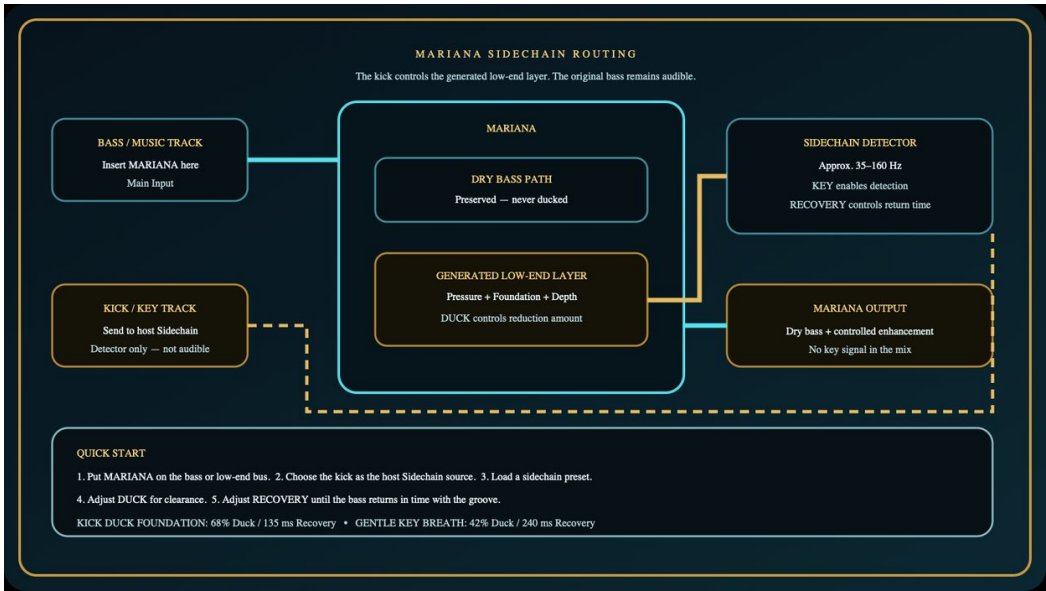
MARIANA exposes an optional stereo host Sidechain bus. Its 35-160 Hz detector uses a 5 ms attack and user-controlled 40-420 ms recovery; the external key is never mixed into the output.

Reduction affects only MARIANA's enhancement delta, preserving dry bass. KEY off is sample-path neutral. The original 14 programs keep KEY off; KICK DUCK FOUNDATION loads 68 percent / 135 ms and GENTLE KEY BREATH loads 42 percent / 240 ms.

Factory programs

MARIANA SWEET SPOT; LOW GRAVITY CLEAN; WARM SUBFIELD; SUB WEIGHT; DEEP PRESSURE; ABYSSAL FIELD; TRENCH FLOOR; GENTLE FINISH; OCEAN FLOOR; SUB ENHANCED; KICK PRESSURE; ABYSS BLOOM; TIDAL TRACE; SHALLOW CURRENT; KICK DUCK FOUNDATION; GENTLE KEY BREATH

Layer-safe sidechain routing



MONOLITH

Low-end impact and body shaper with excavation, weight, resonance, reverse motion, drive, balance, and an optional kick-key sidechain that clears only the added stone/weight layer.

Declared product: MONOLITH | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Low End Source | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
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Parameter surface

ID	Display name	Type
gate	gate	Referenced
input	input	Referenced
output	output	Referenced
channelmode	channelmode	Referenced
gainmatch	gainmatch	Referenced
excavation	excavation	Referenced
impact	impact	Referenced
weight	weight	Referenced
resonance	resonance	Referenced
reverse	reverse	Referenced
drive	drive	Referenced
balance	balance	Referenced
sidechain_enable	Sidechain Key	Bool
sidechain_duck	Stone Duck	Float
sidechain_recovery	Key Recovery	Float

Artwork authority

source/monolith_bg.png

Meter and visualizer semantics

Display	What it measures	How to read it
INPUT / OUTPUT	Live pre/post level	Use for bypass and level matching.
IMPACT	Low-end transient activity	Shows the punch component being produced.
WEIGHT	Sustained low-frequency body	Shows foundation rather than peak attack.
RESONANCE	Active resonant low-end energy	Persistent high readings warn of ringing.
REVERSE	Reverse-motion contribution	Shows how strongly the reverse path is active.
Monolith field	Combined low-end activity map	Reactive summary, not a spectrum analyzer.
KEY	Detected 40-180 Hz sidechain energy	Moves only when KEY is enabled and a host sidechain is routed.
STONE DUCK	Reduction of the added stone/weight layer	The original input impact remains outside this reduction.

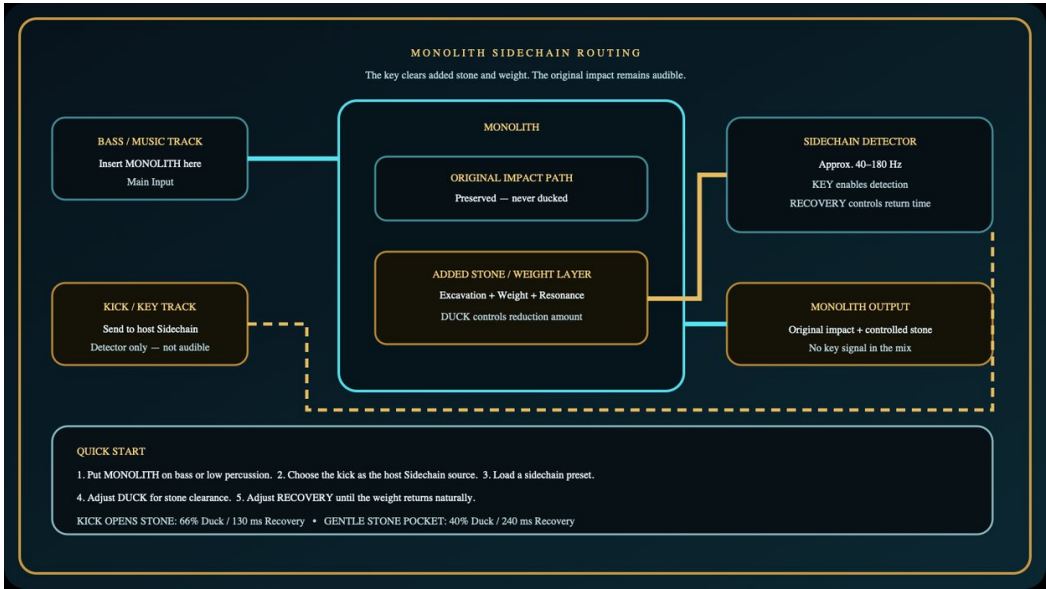
MONOLITH exposes an optional stereo host Sidechain bus. Its detector is band-limited to approximately 40-180 Hz with a 5 ms attack and user-controlled 40-420 ms recovery.

Reduction is applied to the exact wet-minus-dry stone/weight delta before recombination, preserving the original input impact. KEY off is sample-path neutral and adds no lookahead or reported latency.

Factory programs

MONOLITH PHI IMPACT; MONOLITH TRANSIENT STONE; MONOLITH BODY WEIGHT; MONOLITH SHAPE IMPACT; MONOLITH BUS CARVE; MONOLITH GATEWAY; KICK OPENS STONE [Stereo]; GENTLE STONE POCKET [Stereo]

Layer-safe sidechain routing



Depth Boost Pro

Focused low-end depth extension built on delay-based mid/side field placement. Primary role is low-end depth and width; secondarily it is a spatial gain stage, with a dedicated Spatial Gain control (0 dB down to -12 dB) applied after the depth field, plus protected input trim.

Declared product: DEPTH_BOOST_PRO | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Low End Source | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
- outputCeiling call detected
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Parameter surface

ID	Display name	Type
input	Input Trim	Float
gainmatch	Gain Match	Bool
depth	Spatial Depth	Float
output	Spatial Gain	Float
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice

Artwork authority

Assets/Images/Boost.png

Meter and visualizer semantics

Display	What it measures	How to read it
Touch sphere	Depth field plus live level	Water motion follows field/level activity; sphere position remains the output control.
Spatial field	Depth, field energy, and motion	A reactive placement view, not a calibrated spectrum.

NTRPL

Low-end transformation engine for rewrite map, novelty, interval motion, contour, shape, space, originality guard, and an optional kick-key sidechain that clears only the generated rewrite layer.

Declared product: NTRPL | **Version:** 3.22 | **Customer formats:** macOS: Audio Unit (AU) and VST3; Windows beta: VST3 x64

Placement: Low End Source | **Channel mode:** detected | **Dual-mono link:** detected

Source-review status

- Exact DIHEngine::processSample call detected
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Parameter surface

ID	Display name	Type
power	Power	Bool
gainmatch	Gain Match	Bool
chordMap	Rewrite Map	Choice
interpolate	Novelty	Float
motif	Chord Rewrite	Float
rhythm	Motion	Float
pitch	Interval Distance	Float
melody	Contour	Float
shape	Shape	Float
field	Space	Float
guard	Originality Guard	Float
alternateMix	Mix	Float
output	Output	Float
channelmode	Channel Mode	Choice
dualmonolink	Dual Mono Link	Choice
sidechain_enable	Sidechain Key	Bool
sidechain_duck	Rewrite Duck	Float
sidechain_recovery	Key Recovery	Float

Artwork authority

assets/ntrpl.png

Meter and visualizer semantics

Display	What it measures	How to read it
SOURCE	Original input contribution	Shows how much source survives into the main route.
REWRITE	Generated harmonic-rewrite activity	Shows the newly synthesized low-end material.
ALT OUT	Alternative new-signal output	Always represents the generated signal path.
MAIN	Final blended output	Use against SOURCE for level and blend context.
Harmonic Rewriter	Live input-to-output contour	Shows source energy changing into rewritten energy.
Rewrite Map	Novelty, chord, interval, contour, motion, space, guard, and mix state	A live process map; it is not a frequency analyzer.

Display	What it measures	How to read it
KEY	Detected 35-160 Hz sidechain energy	Moves only when KEY is enabled and a host sidechain is routed.
REWRITE DUCK	Reduction of the generated rewrite layer	The source bass path remains outside this reduction.

NTRPL exposes an optional stereo host Sidechain bus. Its key detector is band-limited to approximately 35-160 Hz with a 5 ms attack and user-controlled 40-420 ms recovery. The key is never mixed into the output.

Sidechain reduction multiplies only the generated rewrite/alternate layer. The source-retained main component stays outside the duck multiplier. KEY off is sample-path neutral and adds no lookahead or reported latency.

Factory programs

KICK CLEARS REWRITE [Stereo]; GENTLE REWRITE POCKET [Stereo]

Layer-safe sidechain routing

